A SIGNED CAUCHY–GREEN FLUX IDENTITY AND LINEAR RESIDUAL LOWER BOUND ON PLANAR CORONA CORRECTIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. For the corona problem on planar domains with N radial slit/annular barriers, we prove an exact signed Cauchy–Green contour identity and show that for the derivative-coupled correction family $a_\nu = z^2 f_\nu'$, even when the radial trace cost is below $1/500{,}000$, the full literal $\bar\partial$-residual norm grows linearly at least as $N/400{,}000$ for all $N \geq 21$.

1. Main Lower Bound

Theorem 1.1. *For every $N \geq 21$ and every cap-feasible density in the derivative-coupled corona correction family $a_\nu(z) = z^2 f_\nu'(z)$ on the N-chamber planar domain, the signed Cauchy–Green interior flux identity forces the complete literal residual to satisfy*

$$\|R_N\|_{L^1} \;\geq\; \frac{N}{400{,}000},$$

excluding uniform boundedness as $N \to \infty$ within that correction ansatz.

Date: September 5, 2026.

2020 *Mathematics Subject Classification.* Primary 30H80; Secondary 30H05, 46J15.